

- For a leaf, the prediction = **majority class in that leaf**.
- **If a leaf is tied** (50/50), break the tie by choosing the **parent node's majority class** (simple and deterministic).

Dataset (quick)

Person	Age	Income	Buys Car
1	22	Low	No
2	25	Medium	No
3	28	Medium	Yes
4	35	High	Yes
5	40	High	Yes
6	30	Low	No

We build **3 trees** (each with a bootstrap sample and one random feature). I'll use the same samples as before.

Useful $\log_2$ numbers (we will reuse):

$$\log_2(0.5) = -1$$

$$\log_2(0.25) = -2$$

$$\log_2(0.75) \approx -0.415037$$

$$\log_2(1/3) \approx -1.5849625$$

$$\log_2(2/3) \approx -0.5849625$$

Entropy formula reminder (simple):

Entropy = $-\sum p \cdot \log_2 p$ (sum over classes).

We compute each p , multiply by $\log_2(p)$, then add and change sign.

Tree 1 — sample: persons **1,3,4,5** — feature **Age**

Sample rows:

- 1 (Age=22, No)
- 3 (Age=28, Yes)
- 4 (Age=35, Yes)
- 5 (Age=40, Yes)

Step A — Parent counts & entropy

- Yes = 3, No = 1, total = 4
- $p(\text{Yes}) = 3/4 = 0.75$
- $p(\text{No}) = 1/4 = 0.25$

Compute parent entropy step-by-step:

1. $\text{term_yes} = -p(\text{Yes}) * \log_2(p(\text{Yes})) = -0.75 * (-0.415037)$
 $= 0.75 * 0.415037 = 0.31127775 \rightarrow \text{round to } 0.311278$
2. $\text{term_no} = -p(\text{No}) * \log_2(p(\text{No})) = -0.25 * (-2)$
 $= 0.25 * 2 = 0.5$

Parent entropy = $0.311278 + 0.5 = \mathbf{0.811278}$ (bits)

Step B — Try split Age ≤ 30

- Left (Age ≤ 30): persons 1(No), 3(Yes) $\rightarrow$ Yes=1, No=1 $\rightarrow$ total 2
 - o $p=0.5$ and $0.5 \rightarrow \text{entropy_left} = -0.5*(-1) - 0.5*(-1) = 0.5 + 0.5 = \mathbf{1.0}$
- Right (Age > 30): persons 4(Yes), 5(Yes) $\rightarrow$ Yes=2, No=0 $\rightarrow$ entropy_right = **0.0**

Weighted entropy after split:

- $\text{weight_left} = 2/4 = 0.5 \rightarrow 0.5 * 1.0 = 0.5$
- $\text{weight_right} = 2/4 = 0.5 \rightarrow 0.5 * 0.0 = 0.0$
Entropy_split = $0.5 + 0.0 = \mathbf{0.5}$

Information gain = parent – split = $0.811278 - 0.5 = \mathbf{0.311278}$

Step C — Leaf labels & tie-break

- Right leaf (Age > 30): pure Yes → **Predict Yes**
- Left leaf (Age ≤ 30): counts Yes=1, No=1 → **tie**
Tie-break rule: use **parent majority** (parent has Yes=3, No=1 → majority = Yes).
So we label left leaf **Yes** (deterministic and simple).

Final Tree 1 (sketch)

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rubyCopy code      [Age ≤ 30?]
      /      \
    Yes      No
    /      \
(1:No,3:Yes) (4:Yes,5:Yes)
Predict: Yes Predict: Yes

```

(So Tree1 predicts **Yes** for both branches under our tie rule.)

Tree 2 — sample: persons **2,3,6,4** — feature **Income**

Sample rows:

- 2 (25, Medium, No)
- 3 (28, Medium, Yes)
- 6 (30, Low, No)
- 4 (35, High, Yes)

Step A — Parent counts & entropy

- Yes = 2, No = 2, total = 4
- $p(\text{Yes})=0.5$, $p(\text{No})=0.5$
- parent entropy = $-0.5*(-1) - 0.5*(-1) = 1.0$

Step B — Split: Income = High? (Right = High, Left = Low or Medium)

- Left (Income ≤ Medium): persons 2(No),3(Yes),6(No) → Yes=1, No=2 → total 3
 - $p(\text{Yes})=1/3 \approx 0.333333$, $p(\text{No})=2/3 \approx 0.666667$

- o $\text{term_yes} = -(1/3) * \log_2(1/3) = -0.333333 * (-1.5849625) = 0.5283208$
- o $\text{term_no} = -(2/3) * \log_2(2/3) = -0.666667 * (-0.5849625) = 0.389975$
- o $\text{entropy_left} = 0.5283208 + 0.389975 \approx \mathbf{0.918296}$
- Right (Income = High): person 4 → Yes=1, No=0 → $\text{entropy_right} = \mathbf{0.0}$

Weighted entropy:

- $\text{weight_left} = 3/4 = 0.75 \rightarrow 0.75 * 0.918296 = 0.688722$
- $\text{weight_right} = 1/4 = 0.25 \rightarrow 0.25 * 0 = 0$
 $\text{Entropy_split} = 0.688722 + 0 = \mathbf{0.688722}$

Information gain = parent – split = $1.0 - 0.688722 = \mathbf{0.311278}$

Step C — Leaf labels

- Right (High): pure Yes → **Yes**
- Left ($\leq$ Medium): Yes=1, No=2 → majority **No** → **No**

Final Tree 2 (sketch)

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rubyCopy code      [Income = High?]
      /      \
    No      Yes
(2:No,3:Yes,6:No) (4:Yes)
Predict: No      Predict: Yes

```

Tree 3 — sample: persons **1,2,3,6** — feature **Age**

Sample rows:

- 1 (22, Low, No)
- 2 (25, Medium, No)
- 3 (28, Medium, Yes)
- 6 (30, Low, No)

Step A — Parent counts & entropy

- Yes = 1, No = 3, total = 4
- $p(\text{Yes})=0.25$, $p(\text{No})=0.75$
- $\text{term_yes} = -0.25 * \log_2(0.25) = -0.25 * (-2) = 0.5$
- $\text{term_no} = -0.75 * \log_2(0.75) = -0.75 * (-0.415037) = 0.311278$
Parent entropy = $0.5 + 0.311278 = \mathbf{0.811278}$

Step B — Split: Age ≤ 27 ?

- Left (Age ≤ 27): persons 1(No),2(No) $\rightarrow$ Yes=0, No=2 $\rightarrow$ entropy_left = 0 (pure No)
- Right (Age > 27): persons 3(Yes),6(No) $\rightarrow$ Yes=1, No=1 $\rightarrow$ entropy_right:
 - $p=0.5$ and $0.5 \rightarrow$ entropy_right = 1.0

Weighted entropy:

- $\text{weight_left} = 2/4 = 0.5 \rightarrow 0.5 * 0 = 0$
- $\text{weight_right} = 2/4 = 0.5 \rightarrow 0.5 * 1.0 = 0.5$
Entropy_split = $0 + 0.5 = \mathbf{0.5}$

Information gain = parent – split = $0.811278 - 0.5 = \mathbf{0.311278}$

Step C — Leaf labels & tie-break

- Left (Age ≤ 27): pure No $\rightarrow$ **Predict No**
- Right (Age > 27): Yes=1, No=1 $\rightarrow$ **tie**. Tie-break rule: use **parent majority**.
Parent majority is No (parent had Yes=1, No=3). So we label the right leaf **No**.

Final Tree 3 (sketch)

pgsqlCopy code [Age ≤ 27 ?]

```
      /      \
    Yes      No
    /        \
(1:No,2:No) (3:Yes,6:No)
Predict: No Predict: No
```

(Both branches here predict **No** under our tie rule.)

Final ensemble — predictions (with our deterministic tie rule)

With the three trees we ended up with:

- **Tree1:** predicts **Yes** for Age ≤ 30 and **Yes** for Age > 30 (so Tree1 always Yes here)
- **Tree2:** Income = High $\rightarrow$ **Yes**; else $\rightarrow$ **No**
- **Tree3:** Age $\leq 27 \rightarrow$ **No**; Age $> 27 \rightarrow$ **No** (so Tree3 always No here)

(This looks extreme because of tiny bootstrap samples; random forests usually produce more varied trees with larger data.)

Example test cases

1. Test A: Age=29, Income=Medium
 - o Tree1 $\rightarrow$ Yes
 - o Tree2 $\rightarrow$ Income Medium $\rightarrow$ No
 - o Tree3 $\rightarrow$ Age $> 27 \rightarrow$ No
Votes: Yes, No, No $\rightarrow$ **Final: No** (2 vs 1)
 2. Test B: Age=36, Income=High
 - o Tree1 $\rightarrow$ Yes
 - o Tree2 $\rightarrow$ High $\rightarrow$ Yes
 - o Tree3 $\rightarrow$ Age $> 27 \rightarrow$ No
Votes: Yes, Yes, No $\rightarrow$ **Final: Yes** (2 vs 1)
 3. Test C: Age=24, Income=Low
 - o Tree1 $\rightarrow$ Age $\leq 30 \rightarrow$ Yes
 - o Tree2 $\rightarrow$ Income Low $\rightarrow$ No
 - o Tree3 $\rightarrow$ Age $\leq 27 \rightarrow$ No
Votes: Yes, No, No $\rightarrow$ **Final: No**
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Short checklist of **why** each decision was made (simple)

- **Why split?** Because splitting reduced entropy (uncertainty). Each chosen split had information gain ≈ 0.311 , so it improved purity.
 - **How label at a leaf?** Majority class in that leaf.
 - **How break ties (50/50)?** We used the simple deterministic rule: **choose the parent node's majority class** — easy to compute and explain.
 - **Why parent-majority?** It's deterministic, reproducible, and uses more local information (the parent describes the local region).
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